

Julker Nayen Rafi

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🌐 LinkedIn 🌐 Website 🎓 Google Scholar

Education

Master of Science in Civil Engineering and Construction

Expected May 2027

Georgia Southern University, Statesboro, Georgia, USA

Supervisor: Dr. Ram Krishna Mazumder

- GPA: **4.00/4.00** — 18 credits completed

Bachelor of Science in Civil & Environmental Engineering

February 2021

Shahjalal University of Science and Technology, Sylhet, Bangladesh

- Overall CGPA: **3.61/4.00** (Grade A–)
- Last two-year GPA: **3.76/4.00** (Grade A)

Research Interests

- Risk-Informed Asset Management
- Life-Cycle Cost Assessment
- Trenchless Technology
- Risk Assessment under Natural Hazards
- Structural Rehabilitation of Pipelines
- Leak Detection using Satellite Imagery

Publications

1. **Rafi, J. N.** and Mazumder, R. K. (2026). “Flood Risk-Informed Life Cycle Cost Analysis of Water Distribution Network.” *ASCE UESI Pipelines 2026*. Published. Paper ID: 264.
2. Alam, S., **Rafi, J. N.**, Mazumder, R. K., and Mahmoud, A. B. (2026). “Risk-Based Asset Management of Stormwater Pipelines.” *ASCE AEI-IRD 2026*. Accepted. Paper ID: 108.
3. **Rafi, J. N.** and Mazumder, R. K. (2026). “A Framework to Integrate Seismic Vulnerability into Asset Management of Water Distribution System.” *EERI 13NCEE 2026*. Accepted. Paper ID: 5011.
4. Ahmed, M., **Rafi, J. N.**, and Hossain, M. D. (2021). “Seismic Analysis of Modified Dual System of Eccentrically Braced Frame–Shear Wall: DDBD Method.” *ICERIE 2021*. Published.

Ongoing Research

1. **Rafi, J. N.** and Mazumder, R. K. (2027). “Flood-Integrated Risk Evaluation for Asset Management of Sarasota Water Distribution System.” *In Preparation*.
Topic: Integrated risk assessment for aging and flooding of water pipelines; asset management of water pipelines under dynamic risk assessment.
2. **Rafi, J. N.** and Mazumder, R. K. (2027). “State-of-the-Art Review: Cyber Physical Risk of Drinking Water System and Mitigation Measures.” *In Preparation*.
Topic: Comprehensive review of cyber-physical threats to drinking water infrastructure, threat modeling using the MITRE ATT&CK framework, and evaluation of risk mitigation and resilience strategies.
3. **Rafi, J. N.** and Mazumder, R. K. (2027). “An Integrated Satellite Imagery and Acoustic Data Driven Framework for Condition Assessment of Water Distribution Networks.” *In Preparation*.
Topic: Data-driven condition assessment framework combining satellite imagery analysis and acoustic sensor data for leak detection and structural health monitoring of water distribution pipelines.

Conference Presentations & Participation

- ASCE UESI Pipelines Conference 2026, Detroit, Michigan – Paper Presentation
- EERI 13NCEE Conference 2026, Portland, Oregon – Paper Presentation

- Georgia Southern Student Scholars Symposium 2026 – Poster Presentation
- AEP-CEC Student Research Symposium 2026 – Poster Presentation
- NASTT 2026 No-Dig Show, Palm Springs, California
- NASTT 2025 Southeast Trenchless Conference, Jacksonville, Florida
- NHERI 2026 Mini Conference – Online

Graduate Coursework

Completed Coursework

- Risk Management
- Research Methodology
- Introduction to Missing Data
- Introduction to Artificial Intelligence
- Introduction to Data Science and ML
- Independent Study

Current Coursework

- Advanced Water and Wastewater Treatment

Professional Experience

Structural Safety Engineer Oct. 2023 – Jul. 2025
RMG Sustainability Council

Structural Engineer Jan. 2023 – Sep. 2023
Design Solution Center

Structural Engineer Jan. 2021 – Dec. 2022
Engineering Development Consultants

Technical Skills

Programming	Python, R, Microsoft SQL Server Management Studio
Water Modeling	EPANET 2.2, EPANET-JS, SWMM
Structural Analysis	ETABS, STAAD.Pro, SAP2000
GIS & Remote Sensing	QGIS, Google Earth Engine (GEE)
Others	Microsoft Office, Autodesk AutoCAD, Bluebeam Revu, Overleaf

Honors, Scholarships & Awards

- Savannah Foods and Industries Engineering Scholarship, 2026
- ASCE Pipelines Conference Scholarship, 2026
- NASTT No-Dig Show Conference Scholarship, 2026
- Georgia Southern University Travel Grant, 2026
- Talent Pool Scholarship, Junior School Certificate Examination, 2010

Professional Training & Workshops

- Get Mapping Quickly with QGIS, Udemy, 2025
- Three-day online training on Steel Building Deficiencies & Retrofitting, 2025
- Four-month training on Data Science and Machine Learning with Python, OSTAD, Bangladesh, 2025
- Structural Engineering Training Program, S.K. Ghosh Associates LLC and International Code Council (ICC), 2022

Professional Affiliations

- Student Member: ASCE, SEI, UESI, NHERI, EERI, NASTT
- Member: Institution of Engineers, Bangladesh (IEB)